Smart Helmet: v1.00 Plan and Wireframes

How our system gets the helmet data and replaces the supplier screens. What we build first and the screens we need.

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01 Where we are today

The helmet is set up by the supplier. When it switches on it asks the supplier's address server in China where to report and then sends everything there. Their control room screens are old and hard to use so we replace them. These are the steps to get the helmet's data to our own server.

Step	What happens	What we need
1	**Now: a pretend helmet.** We build our server and a pretend helmet that sends the same messages as the real one from the Interface Document. We can build and test everything while we wait.	Nothing from the supplier. We can start now.
2	**A real helmet.** The helmet still asks the supplier's address server first. With the admin login we change the two addresses it gets back to ours. Then it reports to our server (Interface Document p3).	A sample helmet and its admin login.
3	**Later: our own address server.** Camera models can be changed to ask our address server instead of the supplier's (Solution 2, pp3 and 4). We build one small page that answers with our two addresses. Nothing then goes to China for the address.	The steps for the two ways to change the address inside the helmet (question 1). After v1.00.

EASY TO CHANGE

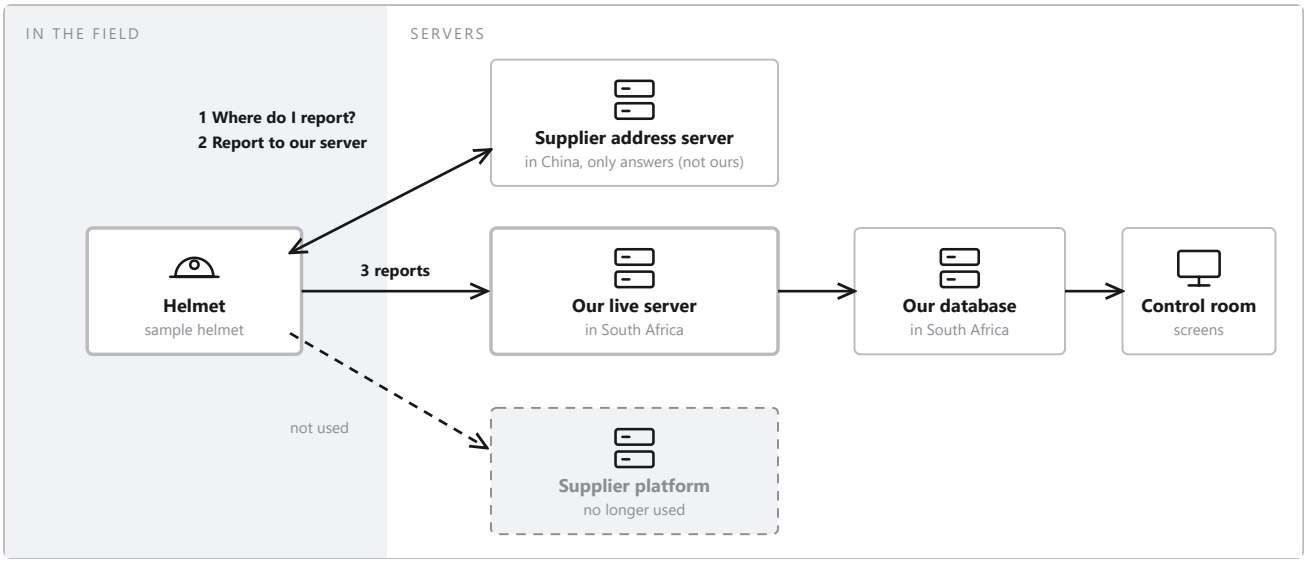
The helmet only needs two addresses. Going from step 2 to step 3 changes where the helmet asks for them. Our server and our screens stay the same. We want camera models and the document says step 3 works on those.

V1.00 IS STEPS 1 AND 2

It proves that our server can receive the helmet and that the control room can see where a worker is. Underground comes in v2.00. Workers also need to be tracked on the surface so this is useful on its own.

02 How our system works in v1.00

The helmet still asks the supplier where to report. The supplier's server only answers with our addresses. The reports go to us.



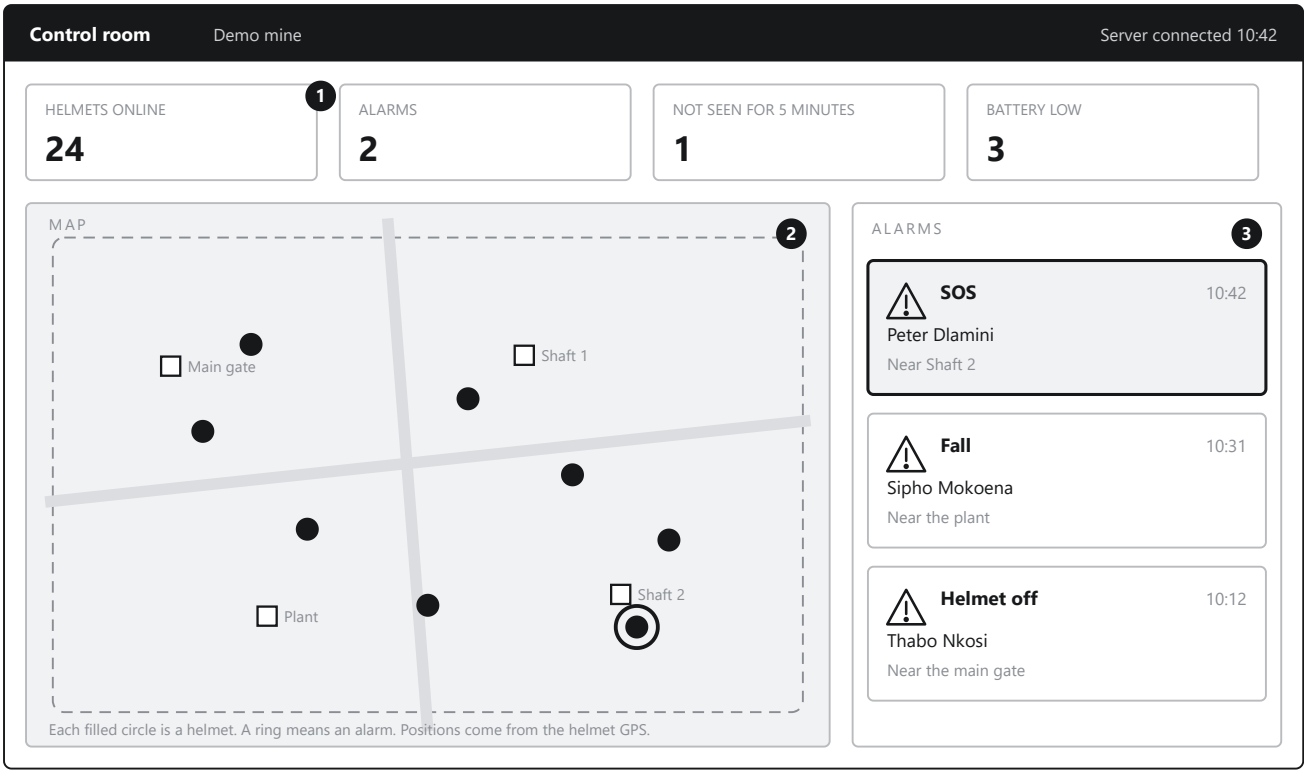
As far as we know nothing else goes to China. We will check this with a network test on the sample helmet.

03 What we build and how we know it works

No.	What we build	How we know it works
1	Our live server. It accepts the helmet's login and its report every 10 seconds.	The pretend helmet connects and its position appears in a log.
2	The database. It keeps every report: position and battery and alarms and nearby Bluetooth devices.	The reports are still there after a restart.
3	The control room screens on the next pages.	The pretend helmet moves on the map.
4	Alarms.	An SOS from the pretend helmet shows on screens 1 and 3.
5	The Bluetooth test list.	A real beacon beside a helmet appears in the list.
6	Connect the real helmet.	Its position matches where it really is.

04 Screen 1: Overview

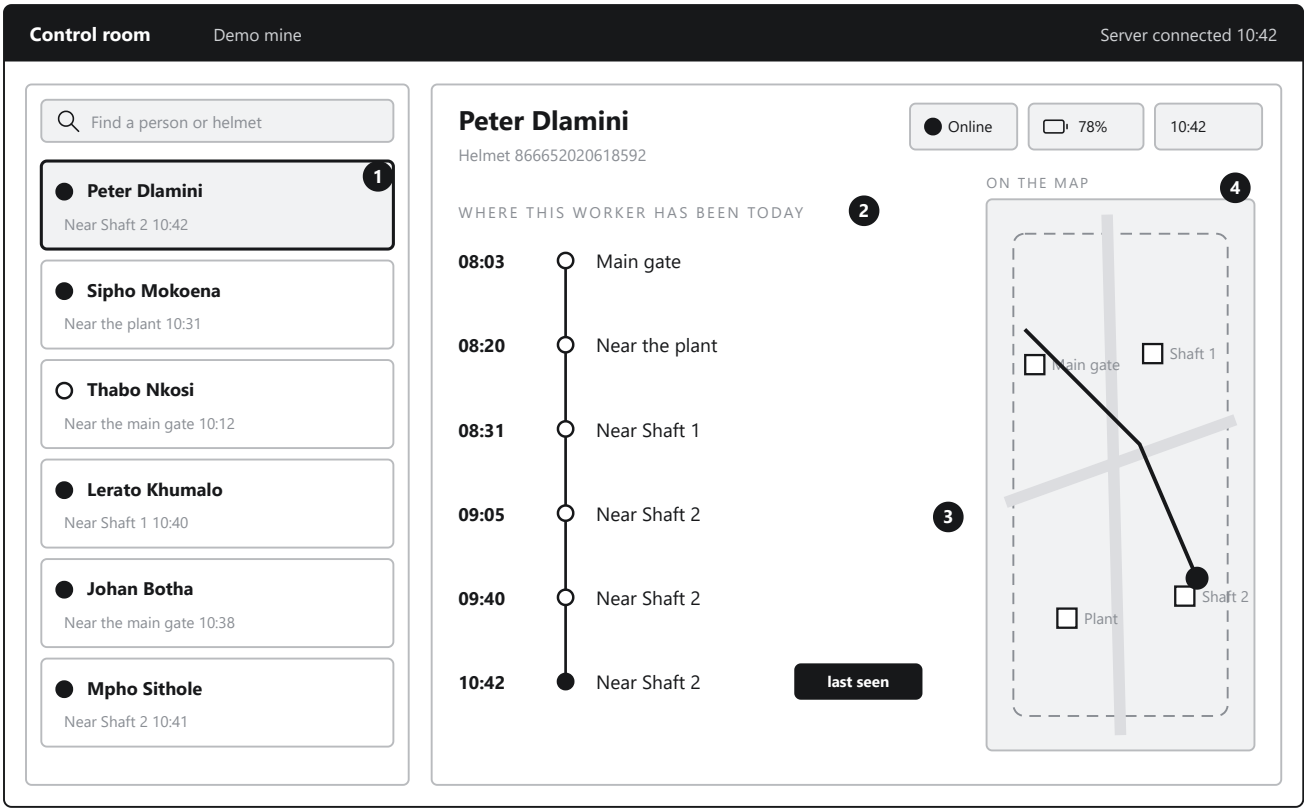
The first thing the control room sees. It answers: is everyone fine and what needs attention?



1. Four numbers at the top. A glance shows if anything is wrong.
2. The map of the site. Each filled circle is a helmet and a ring is an alarm. Positions come from the helmet GPS so this works above ground only.
3. The alarms list. The newest alarm is at the top. Tapping one opens screen 3.

05 Screen 2: One worker

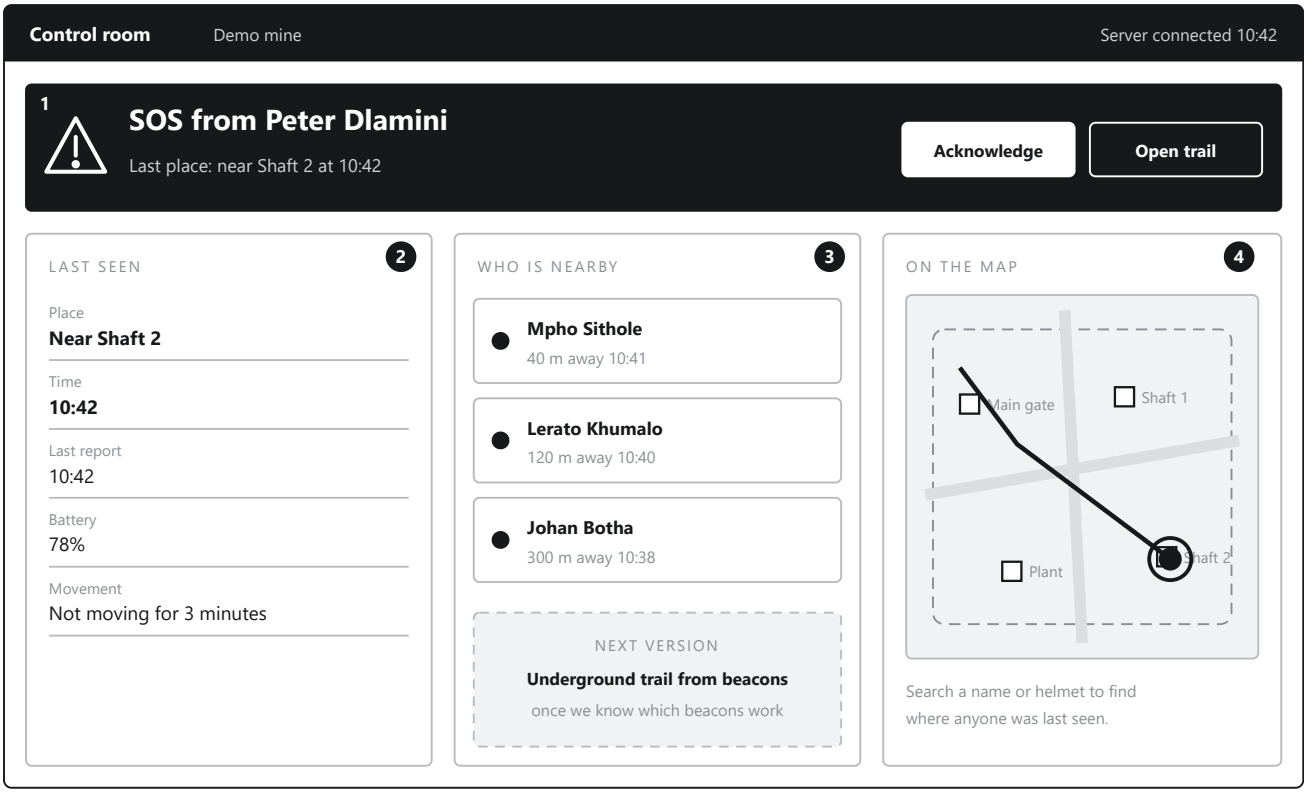
Pick a worker to see where they have been today from the helmet GPS.



1. A list of workers with a search box. A filled dot means the helmet is online. An empty dot means it has not been seen.
2. Where the worker has been today with the time. The last line is where they were last seen.
3. The black tag marks the last time the helmet reported.
4. The same route drawn on the map. The map depends on question 3 because the helmet may report positions in a Chinese map system.

06 Screen 3: Alarm and last seen

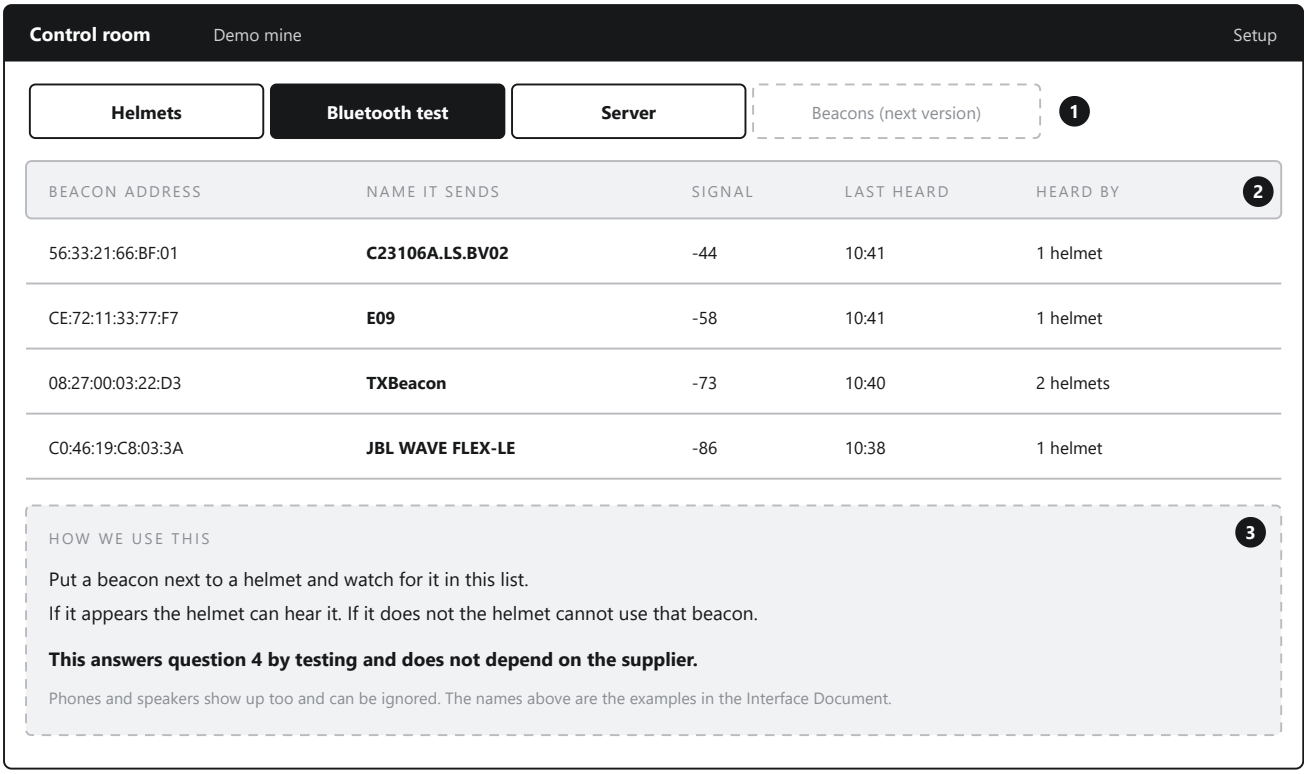
What opens when an alarm comes in. It is also how the control room finds out where someone was last seen.



1. The alarm bar. Who it is and where they were last seen. Two buttons: acknowledge or open the trail.
2. The last seen card. The place and the time and the battery. If a worker stops moving it says for how long.
3. Who is close to the worker by GPS. They can be called first. The dashed box shows what comes in the next version once the beacons are confirmed.
4. The way the worker came on the map. The search finds anyone by name.

07 Screen 4: Setup and Bluetooth test

Where we check which beacons the helmet can hear and where helmets report. This is how we find out what works before we build the underground part.



1. Tabs for helmets and the Bluetooth test and the server. The server tab shows the address the helmets report to, which is our own. The beacons tab is for the next version.
2. Every Bluetooth device the helmets hear with its signal strength. The helmet report lists all of them.
3. How we use it. We put a real beacon next to a helmet and see if it shows up. That tells us which beacons work.

WHAT V1.00 DEPENDS ON

A sample helmet and its admin login on the supplier's platform so we can point it to our server. Question 3 on the map. The underground trail in the next version also depends on question 4 on beacons and the Bluetooth test on screen 4 can answer it. Everything else in this document can be built from the Interface Document.